

Assignment:-4

Task 1:- Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

- Turn off the computer and unplug the power cable.
- Open the computer case/back cover carefully.
- Find the RAM slot on the motherboard.
- Press the clips on both sides of the RAM module.
- Remove the RAM carefully by holding it from the edges.
- Check the RAM and slot for dust, damage, or broken parts.
- Put the RAM back into the slot in the correct direction.
- Press it down gently until the clips lock into place.
- Close the case/back cover and reconnect the power.
- Turn on the computer and check if it starts normally.

Task 2:- Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix. Think about display issues, driver checks, and physical connections.

GPU Troubleshooting Checklist

1. ☐ **Check the display** – Look for a black screen, no display, or screen lines. Try another HDMI/DP cable.
2. ☐ **Check the GPU** – Make sure the GPU is properly fitted in the motherboard slot.
3. ☐ **Check power cables** – Make sure the GPU power cable is connected properly.
4. ☐ **Check the driver** – Update or reinstall the GPU driver.
5. ☐ **Check for dust and heat** – Clean dust and make sure the GPU fan is working.
6. ☐ **Try another monitor** – Connect the PC to another working monitor.
7. ☐ **Confirm the fix** – Start the PC and play a game. If the display works normally without crashes, the problem is fixed.

Task 3:- Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Storage Issue Simulation

1. Turn off the PC and unplug the power.
2. Disconnect the SATA/NVMe drive carefully.
3. Turn on the PC.
4. Before reconnecting, you may see:
 - "No Boot Device Found"
 - "Operating System Not Found"
 - The drive may be missing in BIOS.
5. Turn off the PC again and reconnect the drive properly.
6. Turn on the PC and enter BIOS.
7. After reconnecting, the drive should appear in the BIOS storage list.
8. The PC should then boot normally.

Task 4:- Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

PC Randomly Restarts – Troubleshooting

Component I would check first: RAM

Reason: A loose or faulty RAM can cause random restarts, crashes, or freezing. It is also easy to check.

Diagnostic Steps

1. Turn off the PC and unplug the power.
2. Remove the RAM carefully.
3. Check the RAM for dust or damage.
4. Put the RAM back firmly into the slot.
5. If there are two RAM sticks, test one stick at a time.
6. Start the PC and check if it still restarts.
7. If the problem continues, test the RAM in another slot or with another working RAM stick.